Varicocele: An Endocrinological Perspective.

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Varicocele affects 15% of male population but it is more frequently identified in patients searching medical care for infertility. The impact of varicocele on semen production and fertility is known, but the relationship between clinical varicocele and impaired hormonal production is not clear. In published literature there are some studies regarding hormonal alterations in patients with varicocele but no review in which all the hormonal findings are explained. The aim of this review is to evaluate, by most common search engine, what is known about hormonal alterations in varicocele-bearing patients, to verify if a cause-effect relationship is documented and to give a useful contribution to in clinical management of this kind of patients. We found contradictory results about hormonal status from literature. Some studies confirmed a decrease of testosterone levels and higher FSH and LH levels that normalize after varicocelectomy, others found lower than normal levels of dihydrotestosterone due to decreased activity of epididymal 5-α-reductase. Lower circulating Anti-Müllerian Hormone levels, accompanied by a decreased Inhibin-B level, were reported as indicators of the decreased Sertoli cells function in varicocele-bearing adult patients. The finding of higher basal 17-OH-progesterone concentrations in patients with varicocele was explained by some authors with a testicular C-17,20-lyase deficiency. There is no doubt that varicocele could led to hormonal alterations. This review proposes that the impaired free sexual steroid levels are the result of a slight, deep-rooted defect in the testes of a certain amount of men with varicocele but further multicentre, randomized controlled studies remain mandatory to better clarify the hormonal features of patients with varicocele and to assess the utility of hormonal evaluation for establishing the duration of varicocele and for better identifying patients who need surgical correction.

Male infertility due to testicular disorders.

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Male infertility is defined as the inability to conceive following 1 year of regular unprotected intercourse. It is the causative factor in 50% of couples and a leading indication for assisted reproductive techniques (ART). Testicular failure is the most common cause of male infertility, yet the least studied to date. The review is an evidence-based summary of male infertility due to testicular failure with a focus on etiology, clinical assessment, and current management approaches. PubMed-searched articles and relevant clinical guidelines were reviewed in detail. Spermatogenesis is under multiple levels of regulation and novel molecular diagnostic tests of sperm function (reactive oxidative species and DNA fragmentation) have since been developed, and albeit currently remain as research tools. Several genetic, environmental, and lifestyle factors provoking testicular failure have been elucidated during the last decade; nevertheless, 40% of cases are idiopathic, with novel monogenic genes linked in the etiopathogenesis. Microsurgical testicular sperm extraction (micro-TESE) and hormonal stimulation with gonadotropins, selective estrogen receptor modulators, and aromatase inhibitors are recently developed therapeutic approaches for men with the most severe form of testicular failure, nonobstructive azoospermia. However, high-quality clinical trials data is currently lacking. Male infertility due to testicular failure has traditionally been viewed as unmodifiable. In the absence of effective pharmacological therapies, delivery of lifestyle advice is a potentially important treatment option. Future research efforts are needed to determine unidentified factors causative in "idiopathic" male infertility and long-term follow-up studies of babies conceived through ART.

Varicocele related testicular atrophy and its predictive effect upon fertility.

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Varicoceles are the most common reversible cause of male factor infertility, yet approximately 80% of men with varicoceles are fertile. Therefore, it is unclear whether all adolescents should undergo prophylactic varicocelectomy to prevent future infertility or whether a subgroup of patients who are at increased risk for future infertility can be identified and treated. Testicular size discrepancy or hypotrophy of the testis associated with a unilateral varicocele has been suggested as an indication for prophylactic varicocelectomy in adolescents. We examined 946 men attending a urological clinic for complaints other than infertility to determine whether testis size discrepancy was predictive of infertility in men with left varicoceles. A left varicocele was detected on physical examination in 211 men, of whom 173 (82%) had been able to father children and 38 (18%) had never fathered children. A group of 630 men without palpable varicoceles served as controls, including 528 (84%) with a history of fertility. Testicular size was measured using an orchidometer and the average testicular volume difference was obtained by subtracting left from right testicular volume. The mean testicular volume difference for the fertile men without varicoceles (1.6 +/- 0.3 ml.) was significantly lower than the fertile men with varicoceles (3.1 +/- 0.4 ml.) ($p < 0.05$) and infertile men with varicoceles (2.5 +/- 0.6 ml.) ($p < 0.05$). There was no significant difference between fertile and infertile men with varicoceles. This study confirms prior reports that the majority of men with left varicoceles are able to father children and that varicoceles cause significant ipsilateral testicular atrophy/hypotrophy. However, we were unable to demonstrate a correlation between loss of testicular volume and fertility status in men with left varicoceles. Further study is needed to identify the clinical parameters predictive of future infertility in adolescents with varicoceles.

Bilateral but not unilateral testicular hypotrophy predicts for severe impairment of semen quality in men with varicocele undergoing infertility evaluation.

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Varicocele is a common cause of infertility, and varicocele-associated testicular hypotrophy has been described as a potential cause of decreased semen quality. We investigated the relationship between testicular hypotrophy and poor semen quality in infertile men with varicoceles. We hypothesized that bilateral hypotrophy is required before the semen quality is severely impaired. We retrospectively identified consecutive patients with palpable varicoceles undergoing an infertility evaluation at a single academic center. Each patient was evaluated by the same clinician with history and physical examination. Testicular hypotrophy was defined as a size discrepancy of greater than 3 mL or an absolute size of less than 14 mL. Multivariate logistic regression analysis was used to determine the clinical predictors of total motile sperm count (TMC) of less than 20 million. A total of 245 men with complete data were identified, and 103 men with a TMC of less than 20 million sperm (mean age 36.2 +/- 6.6 years) were compared with 142 men with normal TMCs (mean age 37.1 +/- 6.5 years). On multivariate analysis, men with bilateral hypotrophy were nearly nine times more likely to have a TMC of less than 20 million sperm than were men without hypotrophy (odds ratio 8.8, 95% confidence interval 2.4 to 32.1), and six times more likely than those with unilateral hypotrophy (odds ratio 6.0, 95% confidence interval 1.4 to 26.3). Unilateral hypotrophy alone did not predict for a low TMC. Among men with varicoceles undergoing infertility evaluation, those with bilateral hypotrophy are at the greatest risk of impaired semen quality.

Advances in surgical treatment of male infertility.

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A male factor is the only cause of infertility in 30% to 40% of couples. Most causes of male infertility are treatable, and the goal of many treatments is to restore the ability to conceive naturally. Varicoceles are present in 15% of the normal male population and in approximately 40% of men with infertility. Varicocele is the most common cause of male infertility that can be corrected surgically. In males with azoospermia, the most common cause is post-vasectomy status. Approximately 6% of males who undergo vasectomy eventually seek reversal surgery. Success of vasectomy reversal decreases with the number of years between vasectomy and vasovasostomy. Other causes of obstructive azoospermia include epididymal, vasal or ejaculatory duct abnormalities. Epididymal obstruction is the most common cause of obstructive azoospermia. Patients with epididymal obstruction without other anatomical abnormalities can be considered as candidates for vasoepididymostomy. With microsurgical techniques, success of patency restoration can reach 70~90%. In case of surgically uncorrectable obstructive azoospermia, sperm extraction or aspiration for in vitro fertilization is needed. Nonobstructive azoospermia is the most challenging type of male infertility. However, microsurgical testicular sperm extraction may be an effective method for nonobstructive azoospermia patients.

Varicocele--the most common cause of male factor infertility?

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Varicocele is often cited as the most common cause of male factor infertility. Arguments in support of this statement include reports of increased prevalence of varicocele in populations of infertile men compared with fertile or otherwise unselected men, association of varicocele with abnormal semen parameters, and improvements in semen parameters and/or pregnancy rates after varicocele repair. Logically, there would appear to be three possibilities regarding the relationship between varicocele and fertility: (i) varicocele has no association with or effect on male fertility; (ii) varicocele may be associated with, but is not the cause of, male subfertility; and (iii) varicocele is a direct cause of male subfertility. In the following, we review evidence from the literature for and against these three possibilities: at the current time, available evidence appears inadequate to confirm or deny any of these three possibilities. Since the ultimate goal of infertile couples is to conceive, it seem logical that future varicocele research should focus primarily on adequately powered, controlled clinical trials in well-characterized infertile couples, randomized to intervention or appropriate controlled observation, with pregnancy as the primary outcome.

Evaluation of the influence of subinguinal varicocelectomy procedure on seminal parameters, reproductive hormones and testosterone/estradiol ratio.

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Varicocele is the most commonly surgically curable cause of male infertility. However, the mechanisms related to the effect of reducing fertility potential have not been clearly identified. The aim of this study was to investigate the effects of varicocelectomy on semen parameters, reproductive hormones and testosterone / estradiol ratio. Matherial and methods: Fifty seven patients outcomes were evaluated before and 6 months after subinguinal microsurgical varicocelectomy. Semen parameters, reproductice hormones and testosteron/estradiol ratio results of patients were compared retrospectively. The mean age was 26.8 years. Fifty four (94.7%) patients had grade 3 and 3 (5.3%) patients had grade 2 varicocele. There was a significant increase in semen parameters except semen volume. There was a statistically significant increase in serum testosterone levels, but not on testosterone/ estradiol ratio. According to our results, microsurgical subinguinal varicocelectomy can be recommended for both improving semen parameters and hormonal recovery.

Evaluation of the success rate and complications of conventional varicocelectomy: Do we need microscopic surgery really?

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Varicocele is the most commonly curable cause of infertility in men. Varicocele is found in 15% of the total male population, 35% of men with primary infertility, and 75%-81% of men with secondary infertility. Generally, patients seek microscopic surgery via surfing the Internet, which is not an available option in all medical centers. The purpose of this study was to determine the success rate and complications of conventional varicocelectomy and to compare it with that of the microscopic method. In this descriptive cross-sectional study, 88 patients with varicocele who underwent non-microscopic varicocele surgery in the 15th Khordad Hospital during 2013-2015 were evaluated by the census method. The mean age of patients with varicocele was 27.30 years; 52 patients underwent bilateral varicocelectomy and 36 left varicocelectomy. Surgical complications of non-microscopic varicocelectomy in the studied patients included bleeding and hydrocele formation both in 0.7% and recurrence in 2.8%. Testicular atrophy was not observed in any case. The incidence of recurrence, hydrocele formation, atrophy, and bleeding in non-microscopic varicocelectomy, if done in accordance with its principles, is not more than the microscopic approach and therefore it could be recommended as a safe surgical treatment in centers where microscopic surgery is not available.

Predictors of Semen Parameters Decline Following the Microsurgical Varicocelectomy.

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Objective Varicocele is considered the most common reversible cause of male infertility. However, some men do not clinically improve after surgical repair. We aimed to identify preoperative factors associated with decreased semen parameters and clinical "downgrading" of total motile sperm count (TMSC) following varicocelectomy. Methods We examined men with preoperative laboratory testing and pre- and postoperative semen analyses (SA) who underwent varicocelectomy between 2010 and 2020. Ejaculate volume, sperm motility, sperm concentration, TMSC, and clinical grade of TMSC (in vitro fertilization: 9M sperm) were used to determine postoperative outcomes. Demographic and clinical factors were compared between cohorts. Results Among 101 men who underwent varicocelectomy, 35 (34.7%) had decreased postoperative TMSC with a median follow-up of 6.6 months (interquartile range 3.9-13.6

months). Eleven (10.9%) men experienced TMSC clinical "downgrading" following surgery. Clinical grade III varicocele was significantly associated with decreased sperm motility on postoperative SA (OR 4.1, 95% CI 1.7-10.0, p=0.002), and larger left testicle volume (OR 1.4, 95% CI 1.1-1.8, p=0.02) was associated with clinical "downgrading" after varicocelectomy. Conclusion A small but significant proportion of men experienced a "downgrading" of semen parameters after varicocelectomy. Larger left testis size was associated with clinical downgrading, whereas clinical grade III varicoceles were associated with lower post-treatment sperm motility. These data are critical for preoperative patient counseling.

Varicocele Embolization: Patient Selection: Preprocedure Workup, and Technical Considerations.

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Varicocele refers to an abnormally dilated and tortuous pampiniform venous plexus within the spermatic cord. The prevalence of varicocele is reported to be approximately 15% in the general male population. Its incidence increases with age and has a higher incidence in infertile men. Varicocele treatment (surgical or interventional) is considered one of the most common therapies of reversible infertility in men. Percutaneous embolization offers nonsurgical, minimally invasive option for the treatment of varicoceles, requiring only minimal sedation. In this article, the authors review the clinical and technical details of percutaneous varicocele embolization with a summary of currently available evidence.